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The Editors

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Dear Editors,

We are pleased to submit our manuscript, “**Scaffold-Controlled Evaluation of Molecular Representations for Antimalarial Activity Prediction**”, for consideration as a Research Article in the *Journal of Computer-Aided Molecular Design*.

The manuscript presents a scaffold-controlled comparison of graph, sequence, and topological molecular representations against an ECFP4-random-forest reference, with the molecular benchmark as the primary analysis and the phenotype-only analysis kept separate. We test this question on a frozen panel of 19 836 African antimalarial natural products under random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing GIN, GIN–TFP, GIN–TNE, and ChemBERTa with ECFP4–RF.

Under the scaffold split, the practically relevant scaffold-held-out setting, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); all four learned arms are significantly lower after multiplicity correction. Under the random split the ordering is unchanged: ECFP4–RF reaches 0.943 3, versus 0.912 1 for ChemBERTa, 0.909 8 for GIN, 0.908 4 for GIN–TFP and 0.891 8 for GIN–TNE (all paired $p < 0.000 1$ after Benjamini–Hochberg correction). The comparison shows that the fingerprint baseline remains strongest under scaffold-held-out testing. The topological fusion arm narrows the gap modestly, while projection-weight analysis identifies persistent-image dimensions as a salient block; this is presented as descriptive evidence about representation use, not causal attribution. The fold-independent initialization protocol is specified so that transformer performance is not inflated by cross-fold weight transfer.

As an external transfer analysis, we repeated the ECFP4–RF versus GIN comparison on a molecule-disjoint ChEMBL-derived *P. falciparum* activity panel ($n = 22\,267$ after exclusion of 180 canonical-SMILES overlaps). ECFP4–RF again outperformed GIN under the random (0.954 7 versus 0.923 7) and scaffold (0.919 0 versus 0.884 3) splits. This extends the comparison beyond the internal screening labels, while the ChEMBL labels remain thresholded IC₅₀/EC₅₀ records rather than prospective experiments. The benchmark supports a bounded conclusion: under the evaluated panels, split protocols, and model configurations, the tested learned representations did not improve on ECFP4–RF. We present this as a dataset- and protocol-bounded result, not as a statement about learned molecular models in general. Code, frozen splits and trained checkpoints are available in the public repository (https://github.com/NanaEngo/Malaria_codesV2).

This work is original, is not under consideration elsewhere, all authors have approved the submission, and the authors declare no competing financial interests.

We thank you for your consideration.

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